

# Hemachandrika Rage

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## PROFESSIONAL EXPERIENCE

BI Data Analyst

Dec 2021 – Dec 2022

Tata Consultancy Services, Bengaluru, Karnataka, India

- Designed and managed a daily data ingestion pipeline that reduced operational delays by 15 hours per month by efficiently storing recent data in PostgreSQL on AWS RDS and archiving historical data in AWS S3 for long-term analysis.
- Automated the data pipeline from AWS S3 to Snowflake, enabling both ad-hoc and scheduled loads, which improved analytics performance by 20% and supported strategic insights on multi-year trends in customer preferences and revenue forecasting.
- Integrated Power BI with PostgreSQL and Snowflake using ODBC connectors, eliminating 20 hours of manual work per week, accelerating compliance reporting by 30%, and improving overall operational efficiency.
- Identified seasonal demand spikes by leveraging PostgreSQL for real-time insights and Snowflake for historical analysis, leading to a 15% revenue increase by strategically adding flights on high-demand routes.

Data Analyst Intern

Nov 2020 – Nov 2021

Myzas Technologies PVT LTD

- Automated financial reporting and compliance for an internal audit team at a battery manufacturer using AWS, PostgreSQL, SQL, and Power BI, reducing manual effort by 15 hours per week.
- Developed dashboards in Power BI, improving audit accuracy and reducing reporting errors by 20%.
- Analyzed \$50M+ in purchase-to-payment transactions using SQL, enhancing compliance verification and financial reporting efficiency.
- Conducted cohort analysis on assets using SQL, reducing valuation discrepancies by 3% and improving financial accuracy.

## SKILLS

- Programming Languages** : Python, SQL, R, MySQL, Jupyter Notebook, VS Code, GitHub
- Visualization Tools** : Matplotlib, Seaborn, Excel, Pivot Tables, VBA, Power BI, Tableau,
- Cloud Platform** : Snowflake, AWS, Hadoop, Hive, Spark
- Data Analysis & Modeling** : EDA, Statistical Analysis, Data Cleaning, Data Preprocessing, Pandas, NumPy
- Machine Learning** : Logistic Regression, Unsupervised Learning, Supervised Learning, Predictive Modelling,
- Deep Learning Frameworks** : Keras, TensorFlow, Hypothesis Testing, Regression, BeautifulSoup,
- Natural Language Processing** : NLTK, VADER (Valence Aware Dictionary & sEntiment Reasoner), Latent Dirichlet Allocation.

## PROJECTS

Sentiment Analysis- Customer Feedback Analysis for British Airways [Link](#)

Python, Web Scrapping, NLTK, LDA

Feb 2024 – April 2024

- Implemented and executed Python scripts for web scraping, data preprocessing, and sentiment analysis, improving data retrieval efficiency by 20%.
- Compiled a sentiment analysis report based on 2,500 customer reviews from airlinequality.com regarding British Airways; identified critical service areas for improvement, leading to a noticeable reduction in complaint volume over three months.
- Established detailed visualizations using Python libraries (Matplotlib and Seaborn), which facilitated a 25% faster decision-making process for marketing strategies based on consumer insights.
- Enhanced strategic communication efforts by effectively translating complex data insights into actionable recommendations, leading to a 15% improvement in campaign target alignment based on natural language processing findings.

Neural Network-based Handwritten Digit Recognition [Link](#)

Python, Keras, TensorFlow, CNN

Nov 2023 – Jan 2024

- Constructed a robust pipeline combining convolutional neural networks and classic machine learning techniques for digit recognition; processed 70,000 handwritten samples, enhancing processing speed by 25% while maintaining high accuracy.
- In the beginning, used a multilayer perceptron (MLP) which is also a neural network, and on the test dataset, which helped to achieve an impressive accuracy of over 96%.
- Switched to a CNN architecture with dropout regularization to improve model performance even further and investigate convolutional neural networks (CNNs), which produced even higher accuracy.
- By utilizing methods including data preparation, hyperparameter adjustment, and model assessment, the CNN model has improved to surpass 96.58% test accuracy.

## EDUCATION

- Saint Peter’s University, Jersey City, New Jersey

Master of Science in Data Science – CGPA 3.85/4

May 2023 – Nov 2024
- JNTUA CEA, Anantapur, Andhra Pradesh, India

B. Tech in Mechanical Engineering – CGPA 8.06/10

July 2016 – Nov 2020